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UG Program in Artificial Intelligence and Data Science

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1. Title

Explainable AI in Education: A Lightweight Web Dashboard for Student Academic Performance Prediction and Intervention

2. Abstract

Predicting student academic performance and delivering timely pedagogical interventions is vital for improving retention and preventing course dropouts in modern educational institutions. While contemporary research leverages complex deep learning architectures that achieve high accuracy, these models predominantly operate as opaque black boxes and demand specialized computational infrastructure, creating severe trust and accessibility barriers for non-technical classroom educators. This project proposes an explainable, lightweight decision-support framework that bridges the gap between sophisticated machine learning research and everyday classroom utility. By training highly interpretable supervised machine learning algorithms (Decision Trees, Support Vector Classifiers, and Random Forests) on baseline academic, demographic, and behavioral metrics, the system provides reliable early warnings. An integrated Explainable Artificial Intelligence (XAI) layer powered by SHAP (SHapley Additive exPlanations) quantitatively reveals the underlying risk factors driving each prediction. Delivered through an intuitive web dashboard developed with Streamlit, the platform allows educators to input routine student indicators, receive instant transparent risk assessments, and access automated, rule-based intervention strategies to proactively guide struggling students.

3. Introduction

Educational Data Mining (EDM) and Learning Analytics (LA) have emerged as pivotal disciplines in modern pedagogical research, enabling academic institutions to extract actionable insights from student demographic records, historical grades, and dynamic digital learning environments. A primary objective of EDM is the early identification of at-risk students who are susceptible to academic underperformance or course abandonment. Timely identification allows institutional advisors and classroom teachers to formulate proactive counseling, peer tutoring, and personalized learning pathways before academic failure becomes irreversible. However, a critical dichotomy exists in current educational artificial intelligence research: while recent state-of-the-art studies focus heavily on complex neural architectures to maximize statistical accuracy, they create significant practical friction due to black-box opacity and computational complexity. This manuscript presents an end-to-end framework combining interpretable machine learning, Explainable AI, and an accessible web dashboard to deliver actionable intervention analytics directly into the hands of educators.

4. Literature Review

[01] Pan, J. et al. conducted an exhaustive systematic survey evaluating the development of machine learning and educational data mining methodologies applied to student academic performance forecasting over the past decade. The authors systematically analyzed the critical input features, performance evaluation metrics, open-access institutional datasets, and predictive algorithms deployed across higher education to facilitate early warning mechanisms. By identifying vulnerable and struggling students early, such predictive systems aim to reduce institutional attrition and course dropout rates through timely pedagogical interventions. A distinguishing component of their survey is a comparative benchmarking simulation that evaluated multiple traditional and ensemble classifiers across widely recognized educational datasets to identify optimal models. However, the survey highlights that the academic literature lacks consensus on the single most effective algorithm, while existing high-accuracy models often function as opaque black boxes. This absence of transparency prevents non-technical educators from understanding the specific feature interactions driving predictions without specialized data science expertise.

[02] Jha, M. K. et al. proposed a state-of-the-art framework named XGNN-AP (Explainable Graph Neural Network for Academic Performance) that leverages Graph Convolutional Networks and Graph Attention Networks to model complex inter-student relationships and topological dependencies. The study conducted extensive benchmarking against established machine learning algorithms, including Random Forest, Support Vector Machines, and Logistic Regression. The proposed XGNN-AP framework attained an exceptional 99.71% classification accuracy alongside an optimal 1.00 weighted F1-score in predicting final student outcomes. To address the fundamental interpretability challenges associated with deep graph networks, the framework incorporated post-hoc Explainable Artificial Intelligence methods, specifically SHAP and GNNExplainer, to uncover the primary academic, behavioral, and demographic features influencing predictions. This transparent reasoning supports early academic alert systems and tailored instructional guidance. Nevertheless, the architecture relies on computationally intensive graph convolutions and intricate data engineering pipelines, presenting substantial operational barriers for standard classroom deployment by non-technical teaching faculty.

[03] Kalita, E. et al. formulated an advanced deep learning framework utilizing Bidirectional Long Short-Term Memory (Bi-LSTM) recurrent networks to capture bidirectional temporal dependencies in student academic records and forecast second-term GPAs. The proposed Bi-LSTM model achieved an average accuracy of 88.23%, statistically outperforming state-of-the-art gradient boosting classifiers, including CatBoost, XGBoost, and LightGBM. To mitigate the opaque, black-box nature inherent in recurrent neural network architectures, the authors incorporated SHAP (SHapley Additive exPlanations) to provide local and global feature attribution. This transparent explainability layer clearly revealed how historical indicators, such as secondary school performance and standardized entrance scores, govern downstream academic trajectories, assisting faculty in designing targeted student counseling. Nonetheless, the model exhibits high computational training costs, vulnerability to overfitting on small institutional datasets, and significant structural complexity in real-time execution. This creates a pronounced disconnect between high empirical accuracy and practical

usability in resource-constrained educational environments where lightweight, accessible models are essential.

[04] Angeioplastis, A. et al. investigated the application of Educational Data Mining (EDM) techniques to forecast collegiate academic performance and optimize learning outcomes by mining student interaction telemetry from the Moodle Learning Management System (LMS). The authors examined longitudinal educational data comprising 450 students across nine academic semesters, formulating classification targets into binary pass/fail and multi-tier categories (No Pass, Average, Excellent). Utilizing statistical correlation analysis for optimal feature reduction, the researchers evaluated diverse classifiers—including k-Nearest Neighbors, Decision Trees, Random Forests, and Artificial Neural Networks—achieving robust F1-scores exceeding 0.80. The resulting analytical framework empowers educators to detect at-risk students early and refine instructional curricula to support personalized learning pathways. However, existing advisory systems predominantly analyze student metrics in isolation, failing to account for nuanced, individual behavioral fluctuations over time. This critical limitation restricts their capability to deliver dynamic, highly customized pedagogical interventions that adapt to students' progressive study habits throughout the semester.

[05] Gonzales, W. et al. formulated a hybrid artificial intelligence framework that synergistically integrates conventional machine learning classifiers (Logistic Regression, Support Vector Machines, Random Forest) with deep learning architectures (AlexNet, GRU, BiGRU) to forecast student performance. Operating through a structured six-stage cyclical methodology, the ensemble framework demonstrated superior classification accuracy and sensitivity in identifying vulnerable students compared to standalone baseline models. The system processes multi-dimensional student records to provide institutional administrators and instructors with an automated early detection mechanism to support timely pedagogical intervention strategies. A critical limitation identified in the literature is that predictive modeling research is overwhelmingly concentrated on higher education and university environments. Consequently, there remains a severe deficit of accessible, lightweight artificial intelligence systems tailored specifically for Regular Basic Education (RBE) and secondary school institutions. In these foundational educational settings, computing infrastructure, server resources, and technical machine learning expertise among teachers are heavily constrained, necessitating simplified, intuitive analytical platforms.

[06] Johora, F. T. and Hasan, M. N. proposed an interpretable machine learning methodology designed to predict undergraduate student academic achievement while ensuring algorithmic transparency and trustworthiness in automated decision-making. The authors evaluated several supervised algorithms, including Logistic Regression, Random Forest, Support Vector Machines, and XGBoost, constructing a soft-voting ensemble classifier that achieved an optimal prediction accuracy of 86.38%. To make the ensemble's classifications fully intelligible to educators, the researchers integrated Explainable AI (XAI) frameworks, specifically SHAP and LIME, to delineate exact feature attributions for early risk predictions. The authors demonstrated that conventional high-performing machine learning architectures operate as opaque black boxes, creating a major trust barrier for non-technical educators who must justify academic counseling. Providing transparent, feature-level reasoning is therefore essential to bridge the gap between algorithmic predictions and actionable,

teacher-led classroom interventions, allowing instructors to understand why a student is flagged before initiating corrective pedagogical measures.

[07] Paul, R. et al. introduced an explainable machine learning classification and regression framework aimed at combating student attrition and accurately forecasting first-semester cumulative grade point averages (CGPA). By focusing on early baseline academic indicators, the system identifies at-risk students before irreversible academic failure or course withdrawal occurs. The authors incorporated Explainable AI (XAI) techniques to provide granular explanations of the primary variables influencing dropout vulnerability, thereby empowering institutional academic support teams to deploy highly personalized counseling and remediation resources. However, the study underscores that higher educational institutions continue to face severe logistical and technical hurdles in implementing proactive early-warning mechanisms. Most institutional advisory frameworks remain fundamentally reactive, flagging underperforming students only after midterm examination failures or formal course dropouts have already taken place. This highlights the critical need for proactive early-warning tools deployable at semester onset using initial baseline academic records to enable timely, preventative educational support.

[08] Lu, J. et al. investigated the multi-dimensional factors influencing college students' academic achievement by moving beyond conventional grade transcripts to incorporate psychological, behavioral, and demographic variables. Utilizing comprehensive student surveys, Chi-Square statistical association tests, and machine learning classifiers—including Logistic Regression, Support Vector Classifiers (SVC), Random Forests, and XGBoost—the authors analyzed non-cognitive performance drivers across diverse cohorts. The statistical findings confirmed that psychological factors, particularly student learning motivation and time management self-efficacy, exert a profound impact on final academic standing, with the Support Vector Classifier demonstrating superior stability in processing nuanced behavioral attributes. The authors emphasized that prior educational data mining research has heavily overlooked psychological and environmental factors, relying almost exclusively on static historical GPAs and demographic data. This over-reliance on static academic records fails to capture real-time student effort, daily study discipline, and emotional engagement necessary for holistic, highly accurate academic performance forecasting across dynamic educational environments.

[09] Luo, Z. et al. introduced an advanced educational data mining methodology that fuses multi-dimensional spatio-temporal features to significantly enhance the accuracy of student academic performance prediction. Recognizing that student learning behaviors vary dynamically over time and across different educational settings, the authors developed a multi-dimensional modeling architecture that captures temporal engagement fluctuations alongside physical campus mobility patterns. By integrating these complex temporal-spatial feature interactions, the proposed model achieved superior precision in forecasting longitudinal academic trajectories compared to traditional static classifiers. However, the authors noted that conventional institutional analytics frameworks predominantly treat student indicators as isolated, static cross-sectional records without temporal depth. This critical limitation ignores the time-varying nature of student study habits and evolving classroom participation patterns, resulting in predictive models that cannot adapt to progressive

behavioral changes throughout the duration of an academic semester, thereby limiting the practical precision and diagnostic value of longitudinal performance assessments.

[10] Pires, J. P. J. et al. addressed a critical domain-specific challenge by implementing supervised machine learning techniques to forecast student success rates and failure risks in introductory computer programming courses within STEM curricula. By applying educational data mining algorithms to student code submission logs, syntax error patterns, and assignment completion milestones, the researchers identified key behavioral markers that indicate whether a student will grasp fundamental programming constructs. This early diagnostic capability allows instructors to deliver targeted pedagogical tutoring to prevent early course failure and subsequent academic dropout. The study highlighted that introductory STEM and programming courses exhibit disproportionately high attrition rates compared to general university subjects, creating a strong need for localized, domain-tailored early-warning frameworks. Generalized, university-wide predictive models fail to capture the unique cognitive hurdles and syntactical complexities of programming, underscoring the vital necessity of modular predictive models adaptable to specific discipline-level classroom requirements.

[11] Akbar, S. formulated an accessible educational data mining framework utilizing foundational decision trees and association rule mining to forecast final student academic standings from baseline educational records. By analyzing classroom attendance patterns, demographic profiles, internal test scores, and student interaction logs, the study generated clear, deterministic rule-based pathways to predict final examination outcomes. The explicit, transparent structure of decision tree logic provided instructors with easily interpretable insights into the specific behavioral and academic thresholds determining student success or failure. However, the research identified a substantial barrier in the practical application of educational data mining: the widespread lack of accessible, lightweight data pipelines for non-technical educators. Most educational institutions lack dedicated data engineering infrastructure and specialized programming expertise, leaving classroom teachers without intuitive, end-to-end software tools capable of transforming raw student data into immediate, actionable instructional decisions without encountering severe operational and technical friction in everyday school settings.

[12] Liu, Y. et al. explored supervised machine learning techniques to analyze dynamic student clickstream telemetry generated within digital Learning Management Systems (LMS) to predict final course performance. By capturing granular behavioral interaction logs—including video lecture viewing frequencies, course forum engagement, resource download timeliness, and online quiz participation—the researchers linked continuous digital learning habits directly to final examination grades. The predictive models demonstrated that continuous behavioral engagement logs serve as far more accurate real-time indicators of academic trajectory than static demographic snapshots. Nevertheless, the authors observed that educational institutions continue to severely underutilize dynamic clickstream interaction logs in operational analytics. Most institutions remain reliant on historical grades and static registration profiles, failing to incorporate dynamic behavioral metrics that reflect real-time student effort, daily study discipline, and ongoing course engagement throughout the academic semester to guide proactive, timely instructional interventions and academic counseling.

[13] Feng, G. and Fan, M. investigated data preprocessing pipelines and optimal feature selection strategies to improve the classification accuracy of educational data mining algorithms predicting student academic grades. The authors demonstrated that raw institutional educational datasets are inherently characterized by high feature dimensionality, missing values, severe class imbalance, and significant measurement noise. By implementing rigorous data cleaning protocols, normalization techniques, and optimal feature subset selection, the study achieved highly robust classifications across standard machine learning algorithms while effectively mitigating model overfitting. The authors emphasized that standard educational data mining models frequently suffer from poor generalization across diverse student cohorts due to inadequate data curation. Without meticulous feature selection and comprehensive cleaning, predictive algorithms learn spurious correlations, underscoring the critical necessity of structured preprocessing pipelines to ensure consistent model stability, robustness, and generalizability in real-world educational deployments across varied institutional environments and student demographics.

[14] Zhang, Y. et al. conducted a comprehensive comparative review evaluating the predictive efficacy, computational costs, and implementation complexities of various Educational Data Mining (EDM), machine learning, and deep learning algorithms against academic benchmarks. The authors systematically analyzed the operational trade-offs between model accuracy and computational resource demands across diverse institutional datasets, providing guidance on selecting appropriate predictive methodologies. Consequently, educational institutions and non-technical teachers face immense difficulties in selecting and deploying appropriate algorithms within existing hardware and budget constraints. This highlights the urgent requirement for lightweight, computationally accessible machine learning solutions that balance high predictive accuracy with practical classroom usability, eliminating the need for expensive server hardware and specialized technical maintenance in routine academic operations.

[15] Alshanqiti, A. and Namoun, A. formulated a hybrid predictive architecture integrating multi-label classification and regression modeling to predict granular academic achievement brackets and identify influential performance determinants. Recognizing that student learning outcomes are multi-dimensional, the authors designed a framework that forecasts specific letter grade intervals rather than relying on standard binary pass/fail classifications. This granular modeling approach significantly enhanced prediction specificity, enabling educators to pinpoint the exact academic and behavioral attributes responsible for placing students into specific grade categories. The authors underscored that conventional binary classification models fail to capture the nuanced trajectory of student achievement, completely overlooking borderline and underperforming average students who require subtle academic support. Implementing multi-class, granular performance forecasting is therefore vital for enabling stratified, highly targeted pedagogical intervention strategies that effectively support students across all academic performance tiers before serious academic failure or course dropout occurs.

5. Literature Comparative Study Table:

S. No.	TITLE	AUTHOR	YEAR	METHODOLOGY USED	TECHNOLOGY USED	RESULT/ OUTCOMES	GAP IDENTIFIED
01	Academic Performance Prediction: Survey	Pan, J. et al.	2026	Literature Survey	ML, EDM	Benchmarked ML models	No consensus on best model
02	Explainable ML for Student Prediction	Jha, M. K. et al.	2026	XGNN-AP Framework	GNN, SHAP	99.71% Acc, F1: 1.00	Deep models lack transparency
03	Student Performance via Bi-LSTM	Kalita, E. et al.	2025	Time-Series DL	Bi-LSTM, SHAP	88.23% Accuracy	High compute; opaque network
04	Predicting Performance via EDM	Angeioplastis, A. et al.	2025	Classification	kNN, NN, LMS	F1-Score > 0.80	Ignores student dynamics
05	Hybrid AI for RBE Student Prediction	Gonzales, W. et al.	2025	Hybrid ML-DL	SVM, AlexNet	Superior sensitivity	Few models for basic school
06	Explainable AI for Undergraduates	Johora, F. T. & Hasan, M. N.	2025	Voting Ensemble	XGB, SHAP, LIME	86.38% Accuracy	Black-box trust barrier
07	Dropout & Performance via XAI	Paul, R. et al.	2025	Regression & ML	XAI, ML	Forecasted 1st Sem GPA	Late at-risk detection
08	Non-Cognitive Factors in Performance	Lu, J. et al.	2024	Survey & Stats	SVC, XGBoost	Motivation impacts GPA	Overlooked behavioral traits
09	Spatio-Temporal Performance Model	Luo, Z. et al.	2024	Spatio-Temporal	Multi-Dim EDM	High trajectory precision	Static data over-reliance
10	Intro Programming Course Success	Pires, J. P. J. et al.	2024	Supervised ML	EDM, Python	Forecasted coding risk	Needs domain-tailored tool
11	Performance Prediction via Mining	Akbar, S.	2024	Rule Mining	Decision Tree	Rule-based pathways	Lacks accessible pipeline
12	LMS Clickstream Performance Mining	Liu, Y. et al.	2022	Log Mining	Clickstream, ML	Linked logs to exam marks	Underutilized LMS logs
13	EDM Preprocessing & Feature Selection	Feng, G. & Fan, M.	2022	Data Cleaning	Feature Selection	Mitigated overfitting	Poor data curation/noise
14	EDM Methods Review & Comparison	Zhang, Y. et al.	2021	Comparative Analysis	EDM, ML, DL	Accuracy-compute trade-off	Hardware selection limits
15	Hybrid Regression & Multi-Label Model	Alshanqiti, A. & Namoun, A.	2020	Multi-Label Reg	Multi-Label ML	Granular grade tiers	Binary pass/fail is coarse

6. Problem Statement

Educational institutions encounter severe challenges in identifying at-risk students early enough to deploy effective pedagogical interventions. Existing predictive frameworks predominantly rely on computationally intensive deep learning models that achieve high statistical accuracy but function as opaque black boxes, preventing educators from comprehending the underlying root causes of academic failure [01, 02, 03]. This absence of algorithmic transparency creates a substantial trust barrier for non-technical teachers who require explainable insights to justify targeted counseling [06]. Furthermore, complex educational data mining pipelines demand specialized computing infrastructure that is inaccessible in localized classroom settings [01, 02]. To bridge this gap, this microproject develops a lightweight, transparent student performance prediction system using interpretable machine learning algorithms (such as Decision Trees and Support Vector Classifiers) coupled with Explainable AI (XAI) feature attributions. Integrated into an intuitive web dashboard, the system provides educators with instantaneous risk classifications, root-cause explanations, and actionable intervention recommendations using baseline academic indicators.

7. Proposed Methodology

The proposed system architecture is structured into five cohesive phases designed to deliver an end-to-end, interpretable, and computationally accessible decision-support pipeline for educators. The initial phase focuses on dataset acquisition and structured preprocessing utilizing established open-source educational repositories, such as the UCI Student Performance Dataset and student engagement logs. The preprocessing pipeline conducts missing value imputation, outlier handling, categorical encoding using one-hot and ordinal techniques, and feature normalization. To address the prevalent issue of imbalanced academic risk classes, synthetic oversampling methods such as SMOTE are applied to ensure balanced representation across failing, average, and high-performing student categories without introducing synthetic bias.

In the predictive modeling phase, the framework emphasizes lightweight, transparent supervised machine learning algorithms rather than computationally prohibitive deep neural networks. Candidate classifiers—including Decision Trees, Support Vector Classifiers, Logistic Regression, and Random Forests—are trained and cross-validated using stratified k-fold techniques. Hyperparameter optimization is systematically performed via grid search to balance classification precision and model simplicity. The models are trained on early-semester indicators, including attendance rates, assignment submission frequency, internal test scores, and study hour metrics, enabling reliable early warnings within the initial weeks of an academic term.

The third phase integrates an Explainable Artificial Intelligence (XAI) layer utilizing SHAP (SHapley Additive exPlanations) to eliminate black-box opacity and establish algorithmic transparency. For every individual student classification, the XAI module computes local Shapley values that quantitatively quantify the positive and negative contribution of each feature toward the predicted risk tier. This generates intuitive visual explanations, such as waterfall and force plots, allowing non-technical teachers to immediately understand the exact pedagogical and behavioral drivers—such as insufficient study time or declining test trends—that placed a student in an at-risk category.

To ensure seamless classroom adoption, the fourth phase embeds the trained predictive and explainability models into an intuitive, lightweight web dashboard developed using Streamlit and Python. The user interface abstracts all underlying machine learning complexity, providing educators with interactive sliders and form inputs to enter routine student metrics. Upon parameter entry, the dashboard instantaneously renders the predicted academic risk classification alongside corresponding SHAP visual feature attributions, delivering real-time feedback without requiring server infrastructure or programming expertise.

The final phase incorporates a rule-based pedagogical intervention engine that translates model explanations into actionable academic remedies. When a student is flagged as at-risk, the engine analyzes the dominant negative feature attributions identified by SHAP and automatically maps them to tailored intervention protocols, such as recommending peer tutoring for low test scores or automated mentor alerts for irregular attendance. By closing the loop between early risk detection, transparent feature explanation, and actionable pedagogical support, the system provides educators with a comprehensive decision-support tool to proactively improve student retention and academic success.

8. Technology Used

The implementation utilizes a modular, robust open-source software stack anchored in Python 3.10+, ensuring optimal maintainability, reproducibility, and computational efficiency across standard computing environments without specialized hardware acceleration. For the foundational data engineering and statistical preprocessing pipeline, NumPy and Pandas are utilized for multidimensional array manipulation, tabular restructuring, and missing data imputation, while Scikit-Learn provides robust preprocessing modules for standard scaling, ordinal encoding, and stratified data partitioning.

The machine learning modeling and cross-validation workflows are built entirely on Scikit-Learn, leveraging optimized implementations of Decision Tree Classifiers, Support Vector Classifiers (SVC), Logistic Regression, and Random Forest Ensembles. Class imbalance is handled through Imbalanced-Learn using the Synthetic Minority Over-sampling Technique (SMOTE). For the interpretability architecture, the SHAP (SHapley Additive exPlanations) and LIME libraries are integrated to compute exact Shapley feature attributions and generate visual explanation plots.

The interactive web dashboard is developed using Streamlit, an agile Python-based web framework that allows rapid front-end prototyping, dynamic widget binding, and reactive state management. Data visualization within the dashboard is rendered using Matplotlib, Seaborn, and Plotly, allowing teachers to interactively explore student risk trajectories. The entire application is containerized and deployable on lightweight local servers or cloud hosting platforms without requiring external database dependencies.

9. Result / Decision

The expected outcome of the proposed system is a highly reliable, transparent, and computationally responsive classification platform that accurately stratifies students into distinct risk tiers (High Risk, Moderate/Average, Low Risk). Across benchmark evaluations, the lightweight machine learning models are designed to achieve a target classification accuracy exceeding 85% with balanced Precision, Recall, and F1-scores (>0.82) across all performance categories, specifically prioritizing high Recall on at-risk students to minimize false negatives.

The decision framework establishes a principled trade-off between statistical accuracy and algorithmic interpretability. Rather than prioritizing marginal accuracy gains from opaque neural networks, the decision model selects algorithms that preserve direct explainability through SHAP feature attributions. When an educator inputs student parameters, the system outputs an instantaneous risk assessment alongside top contributing factors, enabling data-informed pedagogical decisions within seconds.

10. Limitations / Future Scope

While the proposed framework addresses key accessibility and interpretability limitations of existing educational AI models, certain constraints remain inherent to the current prototype. The system primarily relies on structured, tabular academic datasets and self-reported behavioral indicators, which may introduce subjective survey biases. Furthermore, the standalone web dashboard requires manual parameter entry by faculty, as it currently lacks direct automated API synchronization with live enterprise Learning Management Systems (LMS) like Moodle or Canvas.

Future research and system development will focus on integrating automated REST API connectors to dynamically ingest real-time student clickstream telemetry and assignment submission logs directly from institutional LMS portals. Additionally, the platform will expand to support longitudinal cross-semester tracking, personalized adaptive learning pathway recommendations, and multilingual dashboard interfaces to broaden adoption across diverse, localized educational institutions.

11. Conclusion

This project bridges the fundamental gap between complex, theoretical AI research and practical classroom decision-making by developing an Explainable AI-powered, lightweight web dashboard for student performance prediction and intervention analytics. By systematically addressing the core limitations identified across 15 contemporary research papers—specifically black-box opacity, high technical complexity, and delayed identification—the proposed architecture prioritizes algorithmic transparency and teacher usability over computational excess. Integrating interpretable machine learning with SHAP feature attributions and automated pedagogical recommendations empowers educators to identify at-risk students early in the semester, understand the root causes of academic struggle, and execute timely, data-informed interventions to foster student retention and academic success.

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